

*Clustering-Based Intrusion Detection: A Comparative Analysis Study of
K-Means & Hierarchical Methods*

GitHub Repository Link: <https://github.com/AI-CS-Y2/SA2-Shuaaa-j.git>

Youtube: <https://www.youtube.com/watch?v=GjOb6HvAbAA>

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Abstract—The rapid evolution of cyber threats necessitates advanced Intrusion Detection Systems (IDS) capable of identifying both known and emerging attacks. This research evaluates the performance of two clustering algorithms, K-Means and Hierarchical Clustering, in network intrusion detection using the CICIDS2017 dataset. The study employs unsupervised clustering metrics (Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index) and supervised classification metrics (Accuracy, Precision, Recall, F1-Score) to assess the algorithms' effectiveness. Results reveal that K-Means significantly outperforms Hierarchical Clustering in key metrics such as recall, accuracy, and F1-Score, making it highly effective for identifying malicious traffic. Hierarchical Clustering demonstrates superior global cluster separation with a higher Calinski-Harabasz Index but struggles with low recall, reducing its reliability for intrusion detection tasks. The findings highlight that K-Means is better suited for IDS applications where high detection accuracy is critical, while Hierarchical Clustering may be advantageous in scenarios requiring clear cluster structures for interpretability. This comparative study underscores the trade-offs between clustering accuracy, computational efficiency, and interpretability, offering insights into the optimal application of these techniques in cybersecurity.

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1. INTRODUCTION

1.1 Background

As cyberattacks become more complex, ensuring network security is essential for businesses, governments, and organizations (Ogundairo and Broklyn, 2024). Intrusion Detection Systems (IDS) are critical in detecting malicious activity by monitoring network traffic for abnormal patterns. Traditional detection methods, which rely on predefined rules or labeled data, are increasingly insufficient to handle modern networks' complexity. To address this challenge, machine learning (ML) techniques, such as clustering algorithms, are being explored to enhance IDS performance and detect previously unknown attacks (Ahmed et al., 2025).

1.2 Justification for Choosing the Algorithm

Clustering algorithms group data based on feature similarities, making them particularly useful in scenarios with limited labeled data. These unsupervised learning techniques can uncover hidden patterns or anomalies in network traffic without prior knowledge of what constitutes malicious behavior (Ali et al., 2024).

Hence, why **K-Means** and **Hierarchical Clustering** were chosen for their ability to analyze the CICIDS2017 data. K-Means is computationally efficient and well-suited for large datasets like CICIDS2017, grouping traffic based on features such as packet size and protocol usage, and flagging anomalies in clusters with distinct characteristics. Hierarchical Clustering, on the other hand, excels in creating a detailed dendrogram, allowing cybersecurity professionals to interpret the relationships between traffic patterns and identify nuanced threats like DDoS attacks or brute-force attempts.

The **CICIDS2017** dataset was chosen because it is a widely recognized benchmark for network intrusion detection, containing a comprehensive range of network traffic data, including various attack types. Its diversity and real-world applicability make it an ideal choice for evaluating the performance of intrusion detection algorithms.

1.3 Statement of the Problem

This study evaluates K-Means and Hierarchical Clustering for network intrusion detection using CICIDS201. The objectives are:

1. Examine the strengths and weaknesses of each clustering technique.
2. Analyze the balance between computational efficiency and model interpretability.
3. Explore potential enhancements and applicability to cybersecurity solutions.

1.4 Scope and Delimitations

This study was limited to the following scope/aspects:

- **Clustering Approaches:** Only unsupervised clustering algorithms are considered, specifically K-Means and Hierarchical Clustering.
- **Performance Metrics:** Algorithm performance is evaluated using clustering metrics and supervised metrics.
- **Dataset Constraints:** The study's reliance on the **CICIDS2017** dataset may limit the findings' generalizability to other network environments or traffic patterns (Lanvin et al., 2023). The research does not account for real-time or dynamic network conditions.

2. LITERATURE REVIEW OF THE PROBLEM DOMAIN

2.1 Overview of Intrusion Detection Systems

Protecting network environments from malicious activity requires the use of Intrusion Detection Systems (IDS). These systems continuously monitor network traffic so that they can identify unusual patterns that may be a sign of an attack such as data breaches (Xie, Dong, and Wang, 2021). IDS can be classified into two broad categories: Host-based Intrusion Detection Systems (HIDS) and Network-based Intrusion Detection Systems (NIDS) (Singh, 2021). NIDS monitors network traffic across devices, ideal for distributed networks.

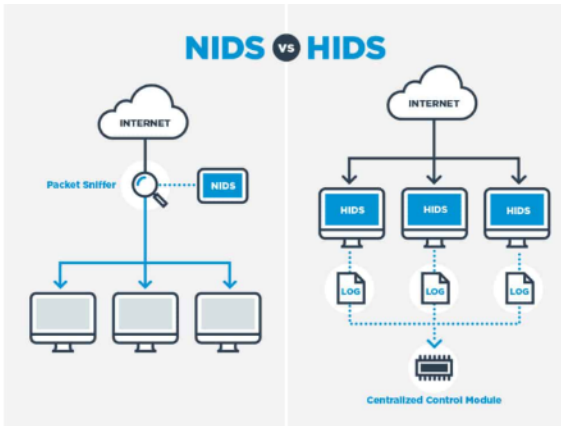


Figure 1: NIDS VS HIDS (Cooper, 2019)

Anomaly-based detection was introduced since traditional IDS solutions struggle to identify new or modified attack vectors (N-able, 2021). Traditional IDS relies on signature-based detection while anomaly-based detection utilizes machine learning and statistical techniques to identify any abnormal patterns in network traffic (Hagar, 2024).

2.2 Clustering Algorithms for Intrusion Detection

One kind of unsupervised machine learning that is frequently used to find

irregularities in networks is clustering algorithms. They function by classifying related data points according to common characteristics, eliminating the need for labels such as "normal" or "attack" (Shifa, 2021). It is simpler to identify outliers or odd patterns that could point to malicious activity thanks to this clustering (Anglen, 2024). Hierarchical clustering and K-Means are two popular clustering techniques for network intrusion detection.

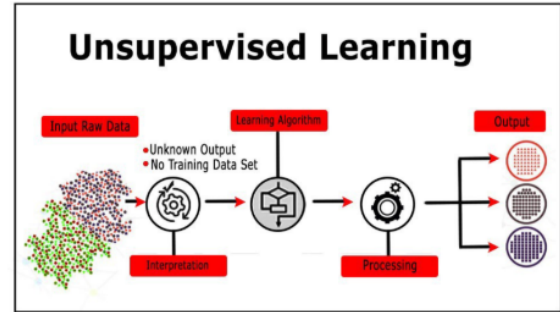


Figure 2: Unsupervised Learning Functions (Kovenko and Shevchuk, 2020)

2.2.1 K-Means Clustering in IDS

K-Means clustering, introduced by Hugo Steinhaus (HolyPython, 2021), partitions data into clusters based on proximity to centroids, adjusting centroids as needed. It's popular in intrusion detection systems (IDS) for its scalability and efficiency (Khedkar and Babhulgaonkar, 2024). However, its performance relies on choosing the right number of clusters, which can be challenging, especially with unknown attack patterns, affecting detection accuracy. K-Means is effective in detecting DDoS attacks.

2.2.2 Hierarchical Clustering in IDS

Hierarchical Clustering organizes data into a hierarchy of clusters, either agglomerative (bottom-up) or divisive (top-down) (Doshi et al., 2020). Hierarchical (Agglomerative) Clustering is commonly used, starting with each data point as its own cluster and merging the closest ones based on a distance metric (Mazarbhuiya,

Alzahrani, and Georgieva, 2020). The process creates a dendrogram, visually representing cluster relationships.

For intrusion detection systems (IDS), hierarchical clustering is valuable for identifying patterns in network traffic. However, its higher computational complexity compared to K-Means can be a limitation for large datasets. Despite this, studies have highlighted its success in detecting intrusions without labeled data (Khraisat et al., 2019).

2.3 Comparison of Clustering Algorithms in IDS

Results from studies comparing clustering algorithms for IDS are not entirely consistent. K-Means and Agglomerative Clustering both did well at detecting intrusions, although K-Means was quicker (Bohara et al., 2020). However, it was emphasized that because of its adaptable structure, hierarchical clustering can identify a greater variety of attack types, even though it requires more computing power.

The requirements of the IDS application frequently determine which of these algorithms is best. For big datasets that need speed and scalability, K-Means is perfect. Hierarchical clustering, on the other hand, works better in situations that require a thorough comprehension of cluster structures, such as locating subgroups within attack patterns (Akram et al., 2021).

3. METHODOLOGY

3.1 Dataset

The study utilizes the **CICIDS2017** dataset (Canadian Institute for Cybersecurity, 2017), a comprehensive benchmark for network intrusion detection with over 2.8 million traffic records, including normal and various malicious activities (e.g., DDoS, DoS, Brute Force). The dataset was preprocessed to handle missing values, scale features, and reduce dimensionality using PCA, yielding a refined dataset for clustering analysis.

3.2 K-Means Clustering

K-Means categorizes network traffic into normal and malicious clusters. Its efficiency in handling large datasets and ability to detect natural patterns make it ideal for distinguishing benign from harmful traffic. The algorithm's scalability and simplicity enhance its effectiveness for network intrusion detection (Dwivedi et al., 2024).

3.3 Hierarchical Clustering

Hierarchical Clustering provides insights into data structure without predefined cluster numbers. Its dendrogram offers a detailed view of relationships between data points, enabling the identification of varying malicious behaviors. The PCA-transformed dataset enhances algorithm performance for this purpose (Mohit Uniyal, 2024).

3.4 Propagation Perspective in Clustering

Although clustering algorithms like K-Means and Hierarchical Clustering don't use neural networks, they share similarities with backward propagation. Both involve iterative refinement to improve outcomes. Clustering uses metrics like Silhouette Score and

Calinski-Harabasz Index, while preprocessing techniques like PCA and feature scaling enhance input quality, akin to optimizing weights in backward propagation, making clustering closer to it than forward propagation (Bergmann and Stryker, 2024)

3.5 Implementation

The code for this project was developed with assistance from **ChatGPT** (OpenAI, 2025), incorporating insights from various resources to enhance its functionality. The implementation of K-Means clustering drew on detailed guidance from **GeeksforGeeks** (2025a), while the methodologies for Hierarchical Clustering were informed by **IBM** research outlined by Noble (2024). Additionally, the creation of an interactive application using Streamlit was supported by tutorials and examples from **GeeksforGeeks** (2025b).

The **CICIDS2017** dataset was processed and analyzed as follows:

(1) Data Preprocessing:

The **CICIDS2017** dataset is preprocessed by cleaning column names, imputing missing values with column means using **SimpleImputer**, and scaling features with **StandardScaler** to enhance clustering performance.

(2) Dimensionality Reduction:

Applied PCA to reduce dimensionality to 10 principal components.

(3) Clustering:

K-Means Clustering & Hierarchical Clustering is applied to the PCA-transformed dataset with the number of clusters set to 2, one for normal traffic and one for malicious traffic.

Additionally, the following **libraries** are used in this clustering analysis:

1. **pandas**: For loading, cleaning, and manipulating the dataset.
2. **numpy**: For numerical computations and handling arrays.
3. **scikit-learn**:
 - **Clustering Algorithms**: Implements KMeans and AgglomerativeClustering for clustering data.
 - **Metrics**: Provides evaluation metrics on the performance metrics scores to assess model performance.
 - **Preprocessing**: Includes tools like StandardScaler for feature scaling and SimpleImputer for imputing missing values.
 - **Decomposition**: Uses PCA for dimensionality reduction to simplify clustering visualization and improve efficiency.
 - **Model Selection**: Offers train_test_split for splitting datasets into training and testing sets.
4. **scipy**: For hierarchical clustering visualizations with dendrogram and linkage.
5. **matplotlib**: For creating plots and visualizations.
6. **seaborn**: For enhancing data visualization with aesthetically pleasing plots.

3.6 Interactive Application on Clustering for Network Intrusion Detection

In this application, you can use two well-known clustering algorithms: **K-Means** and **Hierarchical Clustering** to detect network intrusions. The process uses Streamlit for ease of use, providing both model performance metrics and visualizations for in-depth evaluation (Mhadhbi, 2021).

Step 1: Data Preprocessing

- Upload a CSV dataset.
- Apply PCA to reduce dimensionality and improve efficiency.



Figure 3.1 Interactive Application on Clustering for Network Intrusion Detection

Step 2: Choose Clustering Method

- **K-Means:** Fast, centroid-based clustering with a predefined number of clusters.
- **Hierarchical Clustering:** Builds a hierarchy of clusters for deeper insights.

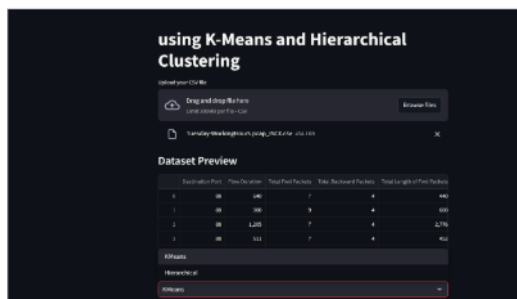


Figure 3.2 Interactive Application on Clustering for Network Intrusion Detection

Step 3: Clustering & Metrics

- Algorithms are evaluated using:
 - **Silhouette Score:** Cluster separation.
 - **Calinski-Harabasz Index:** Compactness and separation.
 - **Davies-Bouldin Index:** Cluster similarity (lower is better).

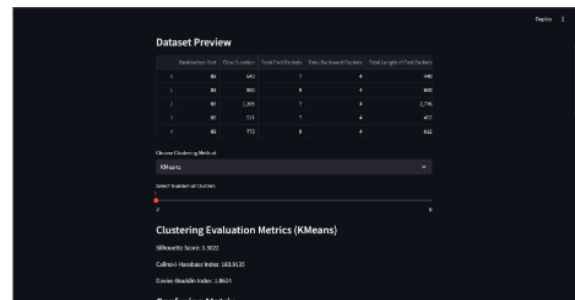


Figure 3.3 Interactive Application on Clustering for Network Intrusion Detection

Step 4: Visualization

- **Confusion Matrix:** Aligns predicted clusters with true labels.
- **PCA Scatter Plot:** Visualizes clusters in 2D for clarity.

By using this interactive application, you can compare the two clustering methods and select the most suitable algorithm for your network intrusion detection task.

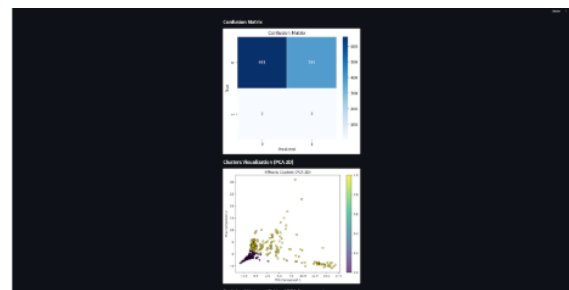


Figure 3.4 Interactive Application on Clustering for Network Intrusion Detection

3.7 Performance Metrics

The following performance metrics were employed:

1. Unsupervised Metrics:

- **Silhouette Score:** Measures how similar an object is to its own cluster compared to other clusters (Sacthesw, 2024).
- **Calinski-Harabasz Index:** Evaluates the dispersion between clusters relative to the dispersion within clusters (GeeksforGeeks, 2024).
- **Davies-Bouldin Index:** Measures the average similarity ratio of each cluster with the one most similar to it (CodeSignal, 2024).

2. Supervised Metrics:

- **Accuracy:** Percentage of correctly classified instances (Alma Better Bytes, 2023).
- **Precision:** Proportion of true positive results out of all positive results predicted by the model (Roberts, 2022).
- **Recall:** Proportion of true positive results out of all actual positive instances (Keldenich, 2021).
- **F1-Score:** The harmonic mean of precision and recall, offering a balance between the two (Sacthesw, 2025).
- **Confusion Matrix:** Showing true positives, true negatives, false positives, and false negatives (IBM, 2024).

4. RESULTS AND ANALYSIS

The results for each model are summarised in the table below:

Metric	K-Means Clustering	Hierarchical Clustering
Silhouette Score	0.9213	0.6173
Calinski-Harabasz Index	250.87	568.47
Davies-Bouldin Index	0.4230	1.6752
Accuracy	0.9817	0.1329
Precision	0.9649	0.9826
Recall	0.9817	0.1329
F1-Score	0.9732	0.2069
Confusion Matrix	[[3434, 2], [62, 0]]	[[403, 3033], [0, 62]]

Figure 4. Table of Summarized Results

4.1 Dendrogram (Hierarchical Clustering)

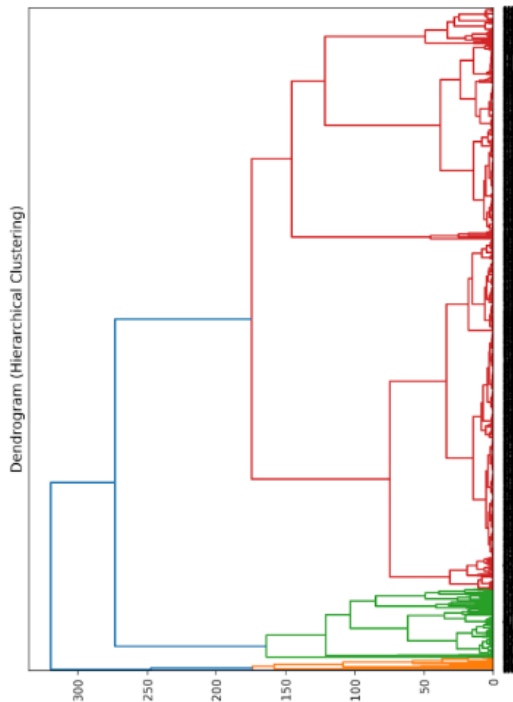


Figure 5. Dendrogram of Hierarchical Clustering

Interpretation: The dendrogram highlights the clustering hierarchy for Hierarchical Clustering. The cutoff point was determined based on the dissimilarity threshold to form 'n' clusters, as shown by the vertical dashed line.

4.2 Cluster Scatter Plot (K-Means and Hierarchical Clustering):

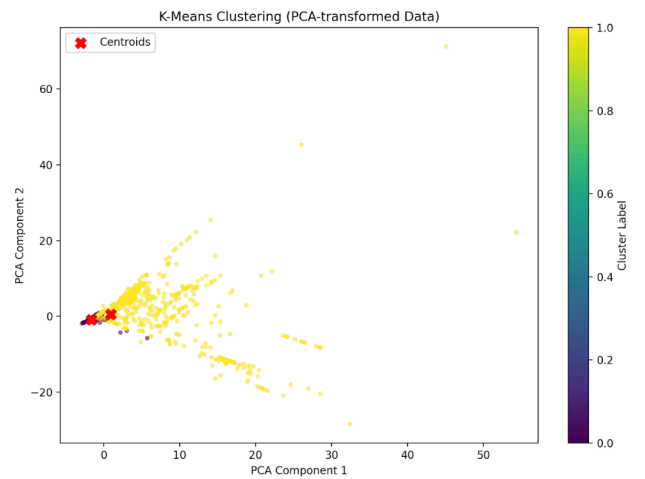


Figure 6.1 Cluster Scatter Plot for K-Means

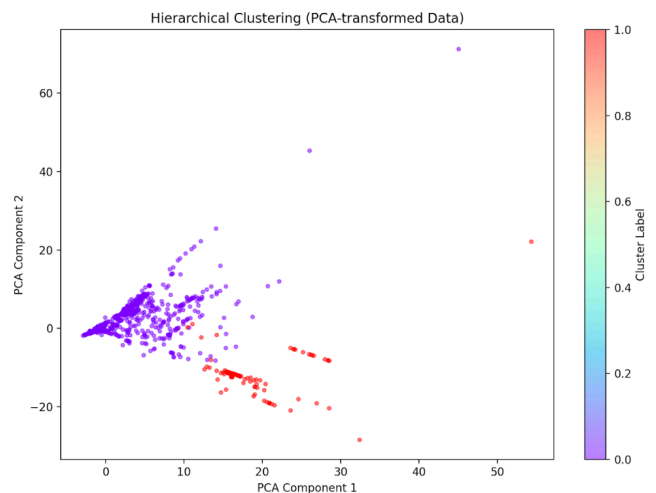


Figure 6.2 Cluster Scatter Plot for Hierarchical Clustering

Interpretation: This plot illustrates the clusters formed by K-Means and Hierarchical Clustering. K-Means displays well-defined clusters with clear centroids, while Hierarchical Clustering shows some overlap between clusters, consistent with its lower Silhouette Score.

4.3 Unsupervised Metrics

4.3.1 Silhouette Score

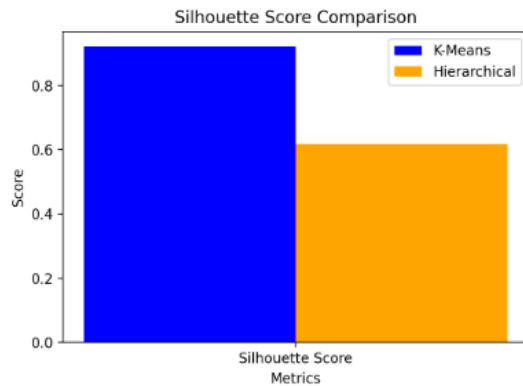


Figure 7. Bar Graph of Silhouette Score of Both Algorithms

Results:

K-Means Clustering: 0.9213

Hierarchical Clustering: 0.6173

Interpretation: K-Means outperforms Hierarchical Clustering in terms of the Silhouette Score, which measures cluster cohesion. With a score of 0.9213, K-Means demonstrates strong internal cohesion and well-defined clusters. In contrast, Hierarchical Clustering has a lower score of 0.6173, indicating less cohesive clusters and more overlap between data points. This suggests that K-Means produces more distinct and well-separated clusters compared to Hierarchical Clustering.

4.3.2 Calinski-Harabasz Index

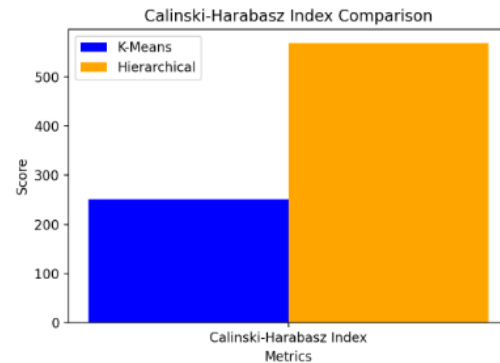


Figure 8. Bar Graph of Callinski-Harabasz Index of Both Algorithms

Results:

K-Means Clustering: 250.87

Hierarchical Clustering: 568.47

Interpretation: The Calinski-Harabasz Index measures the ratio of between-cluster to within-cluster dispersion, with higher values indicating better-defined clusters. K-Means scores 250.87, reflecting a moderate balance between separation and compactness. Hierarchical Clustering, however, scores 568.47, showing superior global cluster separation. This suggests that while K-Means excels in internal cohesion (Silhouette Score), Hierarchical Clustering outperforms overall cluster distinguishability, offering stronger separation despite lower cohesion.

4.3.3 Davies-Bouldin Index

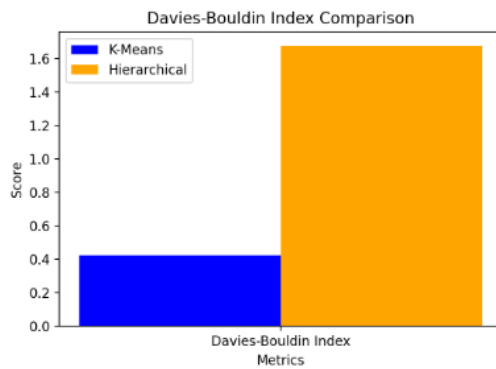


Figure 9. Bar Graph of Davies-Bouldin Index of Both Algorithms

Results:

K-Means Clustering: 0.4230

Hierarchical Clustering: 1.6752

Interpretation: K-Means outperforms Hierarchical Clustering with a much lower Davies-Bouldin Index, indicating highly distinct clusters with minimal overlap. This aligns with its higher Silhouette Score, showing better internal cohesion. In contrast, Hierarchical Clustering has a higher Davies-Bouldin Index, suggesting greater overlap and less distinct clusters.

4.4 Supervised Metrics

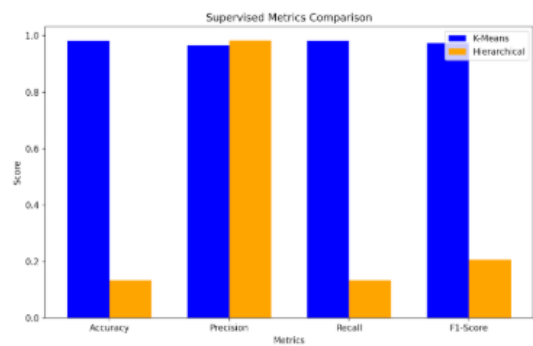


Figure 10. Bar Graph of Supervised Metrics Comparison

4.4.1 Accuracy

Results:

K-Means Clustering: 0.9817

Hierarchical Clustering: 0.1329

Interpretation: K-Means shows excellent accuracy, aligning well with ground truth labels and correctly classifying most instances. In contrast, Hierarchical Clustering has low accuracy, indicating poor alignment with the actual classes, despite forming well-separated clusters. K-Means outperforms Hierarchical Clustering in terms of accurate classification.

4.4.2 Precision

Results:

K-Means Clustering: 0.9649

Hierarchical Clustering: 0.9826

Interpretation: K-Means shows high precision, effectively minimizing false positives, while Hierarchical Clustering slightly outperforms K-Means in precision. However, Hierarchical Clustering's better precision comes at the cost of poor recall. The difference in precision between the two algorithms is minimal, with K-Means remaining highly effective overall.

4.4.3 Recall

Results:

K-Means Clustering: 0.9817

Hierarchical Clustering: 0.1329

Interpretation: K-Means excels in recall, detecting nearly all malicious instances with low false negatives, making it highly reliable for intrusion detection. In contrast, Hierarchical Clustering has very low recall, missing most malicious instances, making it less effective for identifying malicious traffic compared to K-Means.

4.4.4 F1-Score

Results:

K-Means Clustering: 0.9732

Hierarchical Clustering: 0.2069

Interpretation: K-Means achieves a high F1-score, demonstrating a strong balance between precision and recall, making it effective for accurate malicious traffic classification. In contrast, Hierarchical Clustering's low F1-score reflects poor recall, despite high precision, resulting in weaker overall performance compared to K-Means.

4.4.5 Confusion Matrix

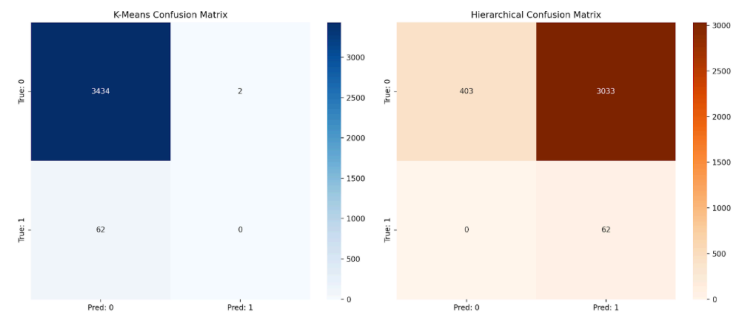


Figure 11. Confusion Matrix of Both Algorithms

Results:

K-Means Clustering: [[3434, 2], [62, 0]]

Hierarchical Clustering: [[403, 3033], [0, 62]]

K-Means has very few false positives and a small number of false negatives, resulting in high accuracy and recall.

Hierarchical Clustering has a high number of false positives, leading to poor accuracy. However, it has no false negatives, which explains its high precision.

K-Means demonstrates a much better balance in the confusion matrix, with fewer errors overall compared to Hierarchical Clustering.

6. SUMMARY OF RESULTS

In this study, two clustering algorithms: K-Means and Hierarchical Clustering were applied to the **CICIDS2017** dataset to evaluate their performance in classifying network traffic and detecting malicious behavior. The evaluation utilized multiple clustering metrics and performance measures.

- (1) **K-Means Clustering** outperforms Hierarchical Clustering in accuracy, precision, recall, F1-Score, and internal cohesion, making it a more reliable option for intrusion detection. However, K-Means occasionally misclassifies malicious instances as benign, which could be a concern in scenarios where detecting all threats is critical.
- (2) **Hierarchical Clustering** demonstrates better global cluster separation with a high Calinski-Harabasz Index, making it useful for understanding overall structure. However, its lower cohesion and weaker classification metrics, including poor recall and numerous false negatives, limit its effectiveness for detecting malicious instances, despite its high precision.

7. CONCLUSION

This study compared K-Means and Hierarchical Clustering for detecting malicious network traffic using the **CICIDS2017** dataset.

The key findings were that K-Means showed a better balance between precision and recall, achieving higher recall and accuracy than Hierarchical Clustering, though its clusters were less distinct. On the other hand, Hierarchical Clustering created well-separated clusters but struggled with detecting malicious instances due to low recall, despite high precision.

K-Means Clustering is ideal for intrusion detection, reliably identifying patterns in network traffic to distinguish between benign and malicious activities. Its high accuracy, precision, and recall ensure minimal false positives, enabling efficient threat detection and response. Scalable and robust, it suits large datasets, enhancing cybersecurity in high-traffic environments like corporate networks and IoT systems.

Hierarchical Clustering, excels in global cluster separation, making it useful for exploratory analysis or applications requiring clear cluster hierarchies. However, its weaker classification metrics and poor recall limit its effectiveness for intrusion detection. Despite high precision, its tendency to misclassify benign instances highlights the need for refinement when applied to cybersecurity tasks.

8. RECOMMENDATIONS

Based on the findings and results of this study, the following recommendations are provided for future research and practical application of clustering algorithms in network intrusion detection:

8.1 For Practical Applications

- **Adopt K-Means for Intrusion Detection:** Given its higher accuracy, recall, and better overall classification performance, K-Means should be considered the primary clustering algorithm for real-time network traffic analysis and intrusion detection systems. Its balance between precision and recall makes it ideal for minimizing false positives and false negatives.
- **Enhance Hierarchical Clustering for Global Separation:** Although Hierarchical Clustering showed weak classification performance, its ability to

produce well-separated clusters can be leveraged in cases where identifying distinct groups in data is more critical than classifying individual instances.

8.2 For Future Research

- **Integrate Supervised Learning with Clustering:** Combining K-Means or Hierarchical Clustering with supervised learning models like Support Vector Machines (SVM) or Random Forests could provide more accurate classifications, as supervised models may offer better generalization for unseen data.
- **Evaluate on More Complex Datasets:** To validate the findings further, testing both K-Means and Hierarchical Clustering on more complex and diverse datasets—such as real-time data from multiple network environments—would be beneficial. This could assess the scalability and robustness of the models when exposed to more dynamic network conditions.

By following these recommendations, future research and applications can further refine the use of clustering algorithms for network intrusion detection and improve overall security outcomes in network systems.

9. REFLECTION

Working on this study has given me a deeper understanding of the strengths and weaknesses of clustering algorithms, especially K-Means and Hierarchical Clustering, in the context of network intrusion detection. I discovered that clustering algorithms can effectively group network traffic, but their performance in classification tasks can vary depending on the algorithm and dataset.

K-Means, which strikes a good balance between precision and recall, seems to be a more practical option for real-world intrusion detection systems, offering a solid solution for detecting malicious activity. On the other hand, Hierarchical Clustering excels at creating well-separated clusters, which can be useful in scenarios where understanding the data's structure and visualizing its relationships are priorities.

This research also demonstrated the potential of AI systems in network intrusion detection, highlighting their ability to identify patterns and classify traffic anomalies. This study emphasized the importance of selecting the right algorithm based on the specific needs and challenges of the task.

10. REFERENCES

- Ahmed, U., Nazir, M., Sarwar, A., Ali, T., Aggoune, E.-H.M., Shahzad, T. and Khan, M.A. (2025). Signature-based intrusion detection using machine learning and deep learning approaches empowered with fuzzy clustering. *Scientific Reports*, 15(1). doi:<https://doi.org/10.1038/s41598-025-85866-7>.
- Akram, R., Ayub, N., Khan, I., Albogamy, F.R., Rukh, G., Khan, S., Shiraz, M. and Rizwan, K. (2021). Towards Big Data Electricity Theft Detection Based on Improved RUSBoost Classifiers in Smart Grid. *Energies*, 14(23), p.8029. doi:<https://doi.org/10.3390/en14238029>.
- Ali, A.H., Charfeddine, M., Ammar, B., Ben Hamed, B., Albalwy, F., Alqarafi, A. and Hussain, A. (2024). Unveiling machine learning strategies and considerations in intrusion detection systems: a comprehensive survey. *Frontiers in computer science*, 6. doi:<https://doi.org/10.3389/fcomp.2024.1387354>.
- Alma Better Bytes (2023). *Metrics for Classification Model*. [online] AlmaBetter. Available at: <https://www.almabetter.com/bytes/tutorials/data-science/classification-metrics>.
- Anglen, J. (2024). *Outlier Detection: What You Need to Know?* [online] Rapidinnovation.io. Available at: <https://www.rapidinnovation.io/post/outlier-detection-what-you-need-to-know>.
- Bergmann, D. and Stryker, C. (2024). *What is Backpropagation?* | IBM. [online] Ibm.com. Available at: <https://www.ibm.com/think/topics/backpropagation>.
- Bohara, B., Bhuyan, J., Wu, F. and Ding, J. (2020). A Survey on the Use of Data Clustering for Intrusion Detection System in Cybersecurity. *International Journal of Network Security & Its Applications*, 12(1), pp.1–18. doi:<https://doi.org/10.5121/ijnsa.2020.12101>.
- Canadian Institute for Cybersecurity (2017). *IDS 2017 | Datasets | Research | Canadian Institute for Cybersecurity* | UNB. [online] Wwww.unb.ca. Available at: <https://www.unb.ca/cic/datasets/ids-2017.html>.
- CodeSignal (2024). *Mastering the Davies-Bouldin Index for Clustering Model Validation*. [online] CodeSignal Learn. Available at: <https://learn.codesignal.com/preview/lessons/1860/mastering-the-daviesbouldin-index-for-clustering-model-validation>.
- Cooper, S. (2019). *10 top network intrusion detection tools for 2018*. [online] Comparitech. Available at: <https://www.comparitech.com/net-admin/network-intrusion-detection-tools/>.

Doshi, I., Sajjalla, S., Choudhari, J., Bhatt, R. and Dasgupta, A. (2020). *Efficient Hierarchical Clustering for Classification and Anomaly Detection*. [online] arXiv.org. Available at: https://arxiv.org/abs/2008.10828?utm_source=chatgpt.com.

Dwivedi, D., Bhushan, A., Singh, A.K. and Snehlata (2024). Leveraging K-means clustering for enhanced detection of network traffic attacks. *2024 3rd International conference on Power Electronics and IoT Applications in Renewable Energy and its Control (PARC)*, [online] (10), pp.72–76.
doi:<https://doi.org/10.1109/parc59193.2024.10486408>.

GeeksforGeeks (2024). *CalinskiHarabasz Index in R*. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/calinski-harabasz-index-in-r/>.

GeeksforGeeks (2025a). *K Means Clustering - Introduction - GeeksforGeeks*. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/k-means-clustering-introduction/>.

GeeksforGeeks (2025b). *A Beginners Guide To Streamlit*. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/a-beginners-guide-to-streamlit/>.

Hagar , S. (2024). *Intrusion Detection Systems: Signature Vs. Anomaly-Based - Cybersecurity Caucus*. [online] Cybersecurity Caucus.

Available at:
<https://cybersecuritycaucus.com/intrusion-detection-systems-signature-vs-anomaly-based/>.

HolyPython (2021). *K-Means History*. [online] HolyPython.com. Available at: <https://holypython.com/k-means/k-means-history/>.

IBM (2024). *Confusion matrix*. [online] Ibm.com. Available at: <https://www.ibm.com/think/topics/confusion-matrix>.

Keldenich, T. (2021). *Recall, Precision, F1 Score - Simple Metric Explanation Machine Learning*. [online] Inside Machine Learning. Available at: <https://inside-machinelearning.com/en/recall-precision-f1-score-simple-metric-explanation-machine-learning/>.

Khedkar, S. and Babhulgaonkar, M. (2024). *Intrusion Detection System using K-Means Clustering and SMOTE*. [online] SlideShare. Available at: <https://www.slideshare.net/slideshow/intrusion-detection-system-using-kmeans-clustering-and-smote/265227243>.

Khraisat, A., Gondal, I., Vamplew, P. and Kamruzzaman, J. (2019). Survey of intrusion detection systems: techniques, datasets and challenges. *Cybersecurity*, [online] 2(1), pp.1–22.
doi:<https://doi.org/10.1186/s42400-019-0038-7>.

Kovenko, V. and Shevchuk, V. (2020). *Clustering with unsupervised learning and key concepts of it — DS/ML course*. [online] machine-learning-and-data-science-with-python.readthedocs.io. Available at: https://machine-learning-and-data-science-with-python.readthedocs.io/en/latest/assignment1_unsup_ml.html.

Lanvin, M., Gimenez, P.-F., Han, Y., Majorczyk, F., Mé, L. and Totel, É. (2023). Errors in the CICIDS2017 Dataset and the Significant Differences in Detection Performances It Makes. *Lecture notes in computer science*, pp.18–33. doi:https://doi.org/10.1007/978-3-031-31108-6_2.

Mazarbhuiya, F.A., Alzahrani, M.Y. and Georgieva, L. (2020). Anomaly Detection Using Agglomerative Hierarchical Clustering Algorithm. *Lecture notes in electrical engineering*, pp.475–484. doi:https://doi.org/10.1007/978-981-13-1056-0_48.

Mehsram, B. and Nalavade, K. (2014). Evaluation of K-Means Clustering for Effective Intrusion Detection and Prevention in Massive Network Traffic Data Kamini Nalavade. *International Journal of Computer Applications*, [online] 96(7), pp.975–8887. Available at: <https://research.ijcaonline.org/volume96/number7/pxc3896526.pdf>.

Mhadhbi, N. (2021). *Python Tutorial: Streamlit*. [online] www.datacamp.com. Available at: <https://www.datacamp.com/tutorial/streamlit>.

Mohit Uniyal (2024). *Hierarchical Clustering in Machine Learning*. [online] Applied AI Blog. Available at: <https://www.appliedaicourse.com/blog/hierarchical-clustering-in-machine-learning/>.

N-able (2021). *Intrusion Detection System (IDS): Signature vs. Anomaly-Based - N-able*. [online] N-able. Available at: https://www.n-able.com/blog/intrusion-detection-system?utm_source=chatgpt.com.

Noble, J. (2024). *What is Hierarchical Clustering? | IBM*. [online] [Ibm.com](https://www.ibm.com). Available at: <https://www.ibm.com/think/topics/hierarchical-clustering>.

Ogundairo, O. and Brooklyn, P. (2024). Machine Learning Algorithms for Intrusion Detection Systems. *Journal of Cyber Security*. [online] Available at: https://www.researchgate.net/publication/382917701_Machine_Learning_Algorithms_for_Intrusion_Detection_Systems.

OpenAI (2025). *ChatGPT*. [online] ChatGPT. Available at: <https://chatgpt.com/>.

Roberts, A. (2022). *Precision: Understanding This Foundational Performance Metric*. [online]

Arize AI. Available at:

<https://arize.com/blog-course/precision-ml/>.

Sacthesw, F. (2024). *Understanding Silhouette Score: A Key Metric for Clustering*. [online]

Future Machine Learning. Available at:

<https://futuremachinelearning.org/understanding-silhouette-score-a-key-metric-for-clustering/>.

Sacthesw, F. (2025). *Understanding the F1 Score: Definition, Importance, Calculation*.

[online] Future Machine Learning. Available at:

<https://futuremachinelearning.org/understanding-the-f1-score-definition-importance-calculation/>.

Shifa, A. (2021). *K-means Clustering Algorithm and Network Intrusion Detection System*.

[online] Medium. Available at:

<https://medium.com/nerd-for-tech/k-means-clustering-algorithm-and-network-intrusion-detection-system-cd7b8de1ed75>.

Singh, H. (2021). *What Is HIDS? HIDS Vs NIDS | HIDS Vs HIPS*. [online] Cyphere. Available at:

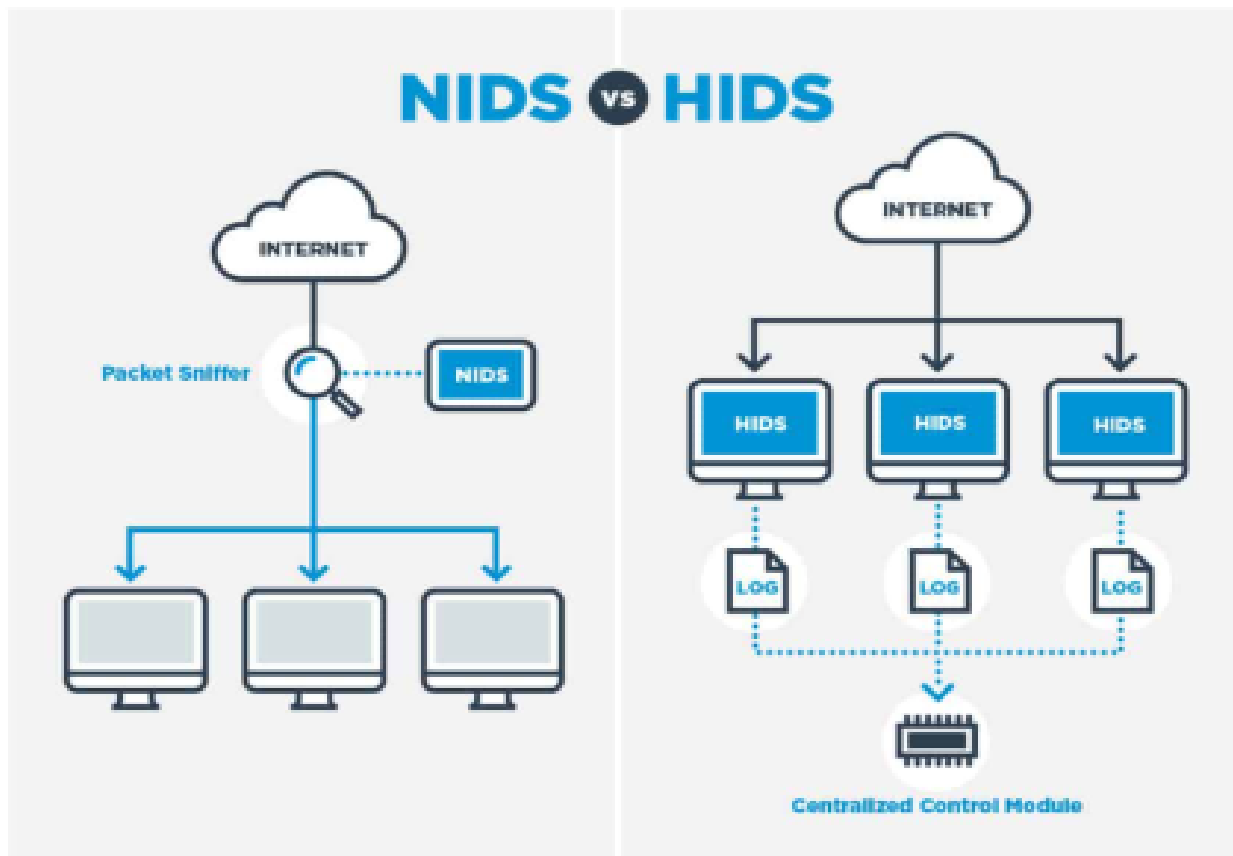
<https://thecyphere.com/blog/host-based-ids/>.

Xie, B., Dong, X. and Wang, C. (2021). An Improved K -Means Clustering Intrusion Detection Algorithm for Wireless Networks Based on Federated Learning. *Wireless communications and mobile computing*, 2021, pp.1–15.

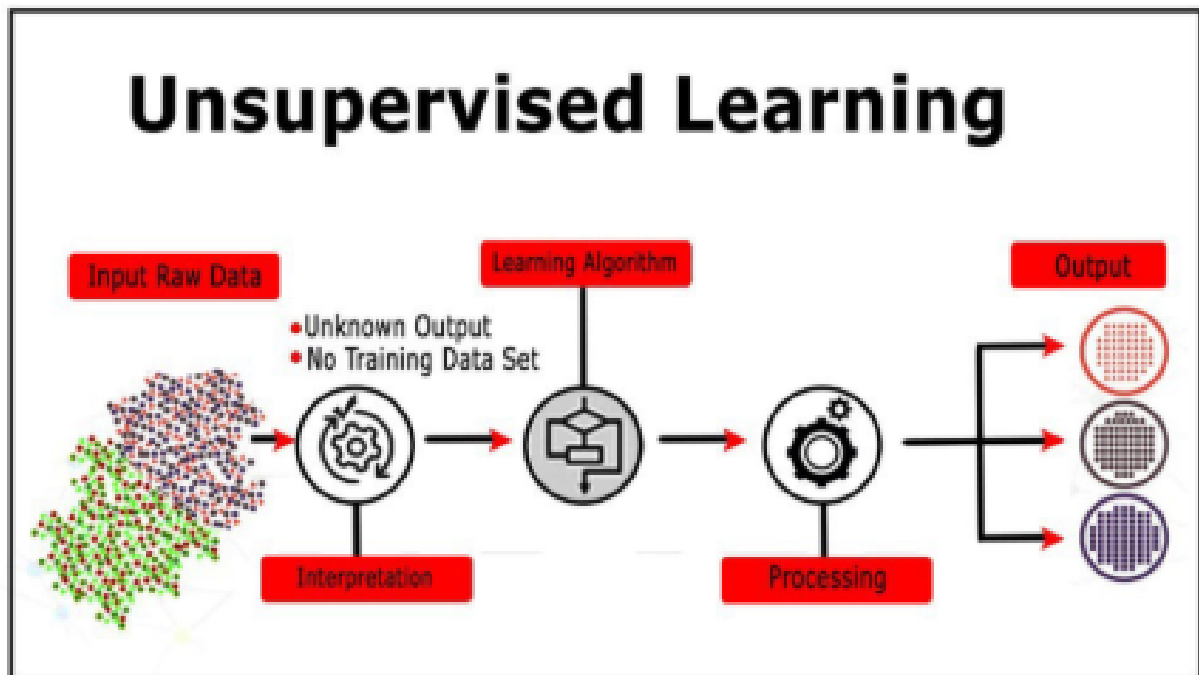
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11. APPENDICES

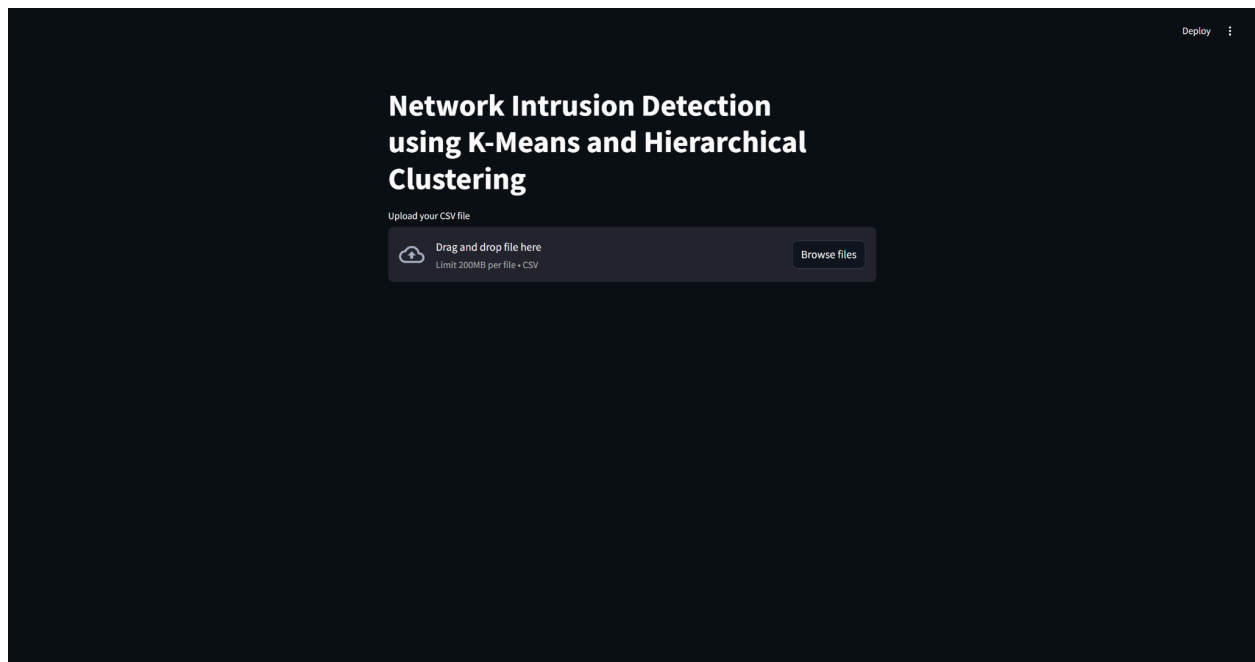
11.1 NIDS VS HIDS



11.2 Unsupervised Learning Functions



11.3.1 Interactive Application on Clustering for Network Intrusion Detection



11.3.2 Interactive Application on Clustering for Network Intrusion Detection

Deploy

Network Intrusion Detection using K-Means and Hierarchical Clustering

Upload your CSV file

 Drag and drop file here
Limit 200MB per file • CSV

Browse files

 Tuesday-WorkingHours.pcap_ISCX.csv 454.1KB

X

Dataset Preview

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets
0	88	640	7	4	440
1	88	900	9	4	600
2	88	1,205	7	4	2,776
3	88	511	7	4	452
4	88	773	9	4	612

Choose Clustering Method

KMeans

KMeans

Hierarchical

2

5

11.3.3 Interactive Application on Clustering for Network Intrusion Detection

Dataset Preview

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Total Length of Fwd Packets
0	88	640	7	4	440
1	88	900	9	4	600
2	88	1,205	7	4	2,776
3	88	511	7	4	452
4	88	773	9	4	612

Choose Clustering Method

KMeans

Select Number of Clusters

2

2

5

Clustering Evaluation Metrics (KMeans)

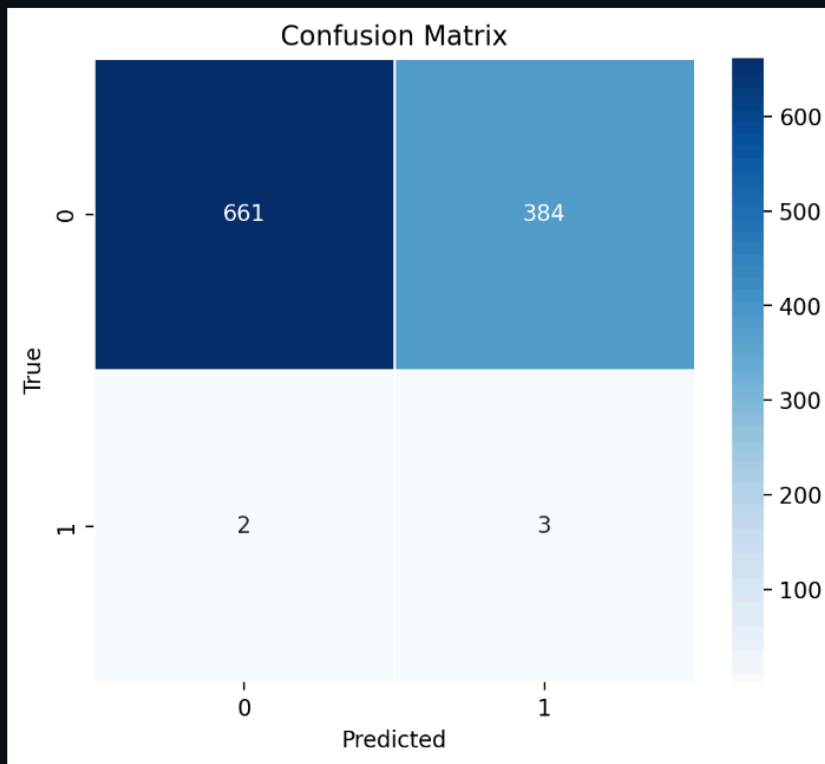
Silhouette Score: 0.3022

Calinski-Harabasz Index: 183.9135

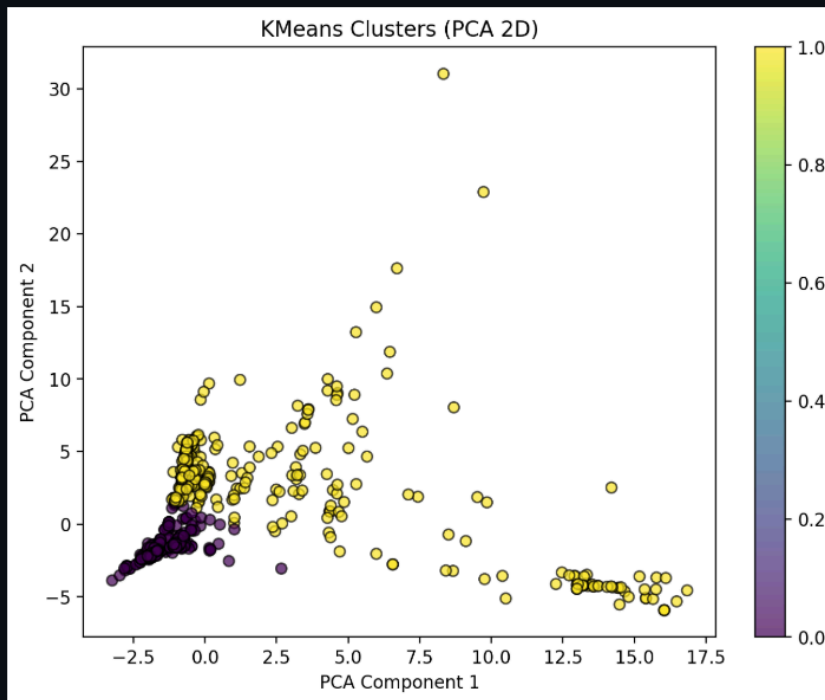
Davies-Bouldin Index: 1.8624

11.3.4 Interactive Application on Clustering for Network Intrusion Detection

Confusion Matrix



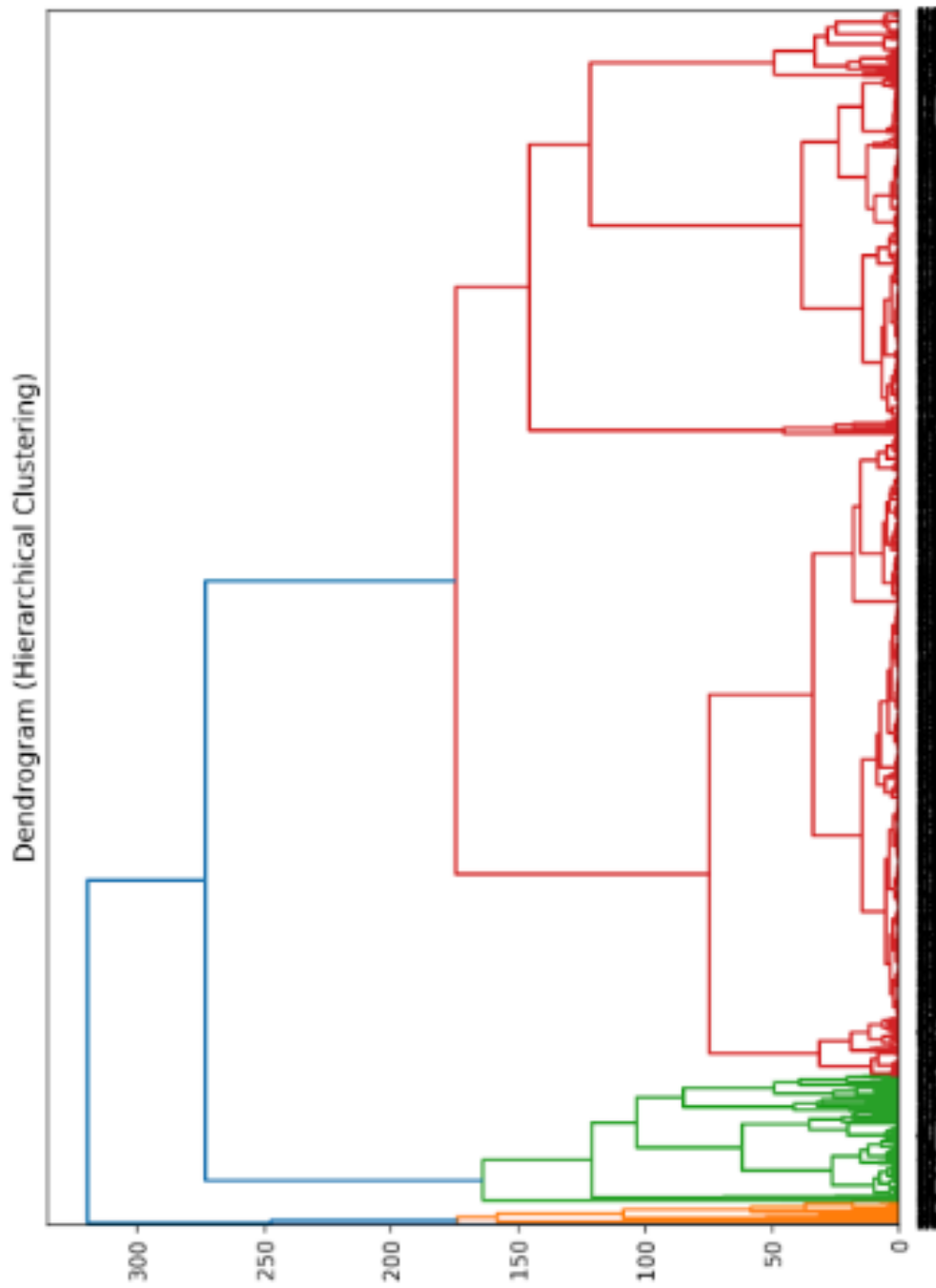
Clusters Visualization (PCA 2D)



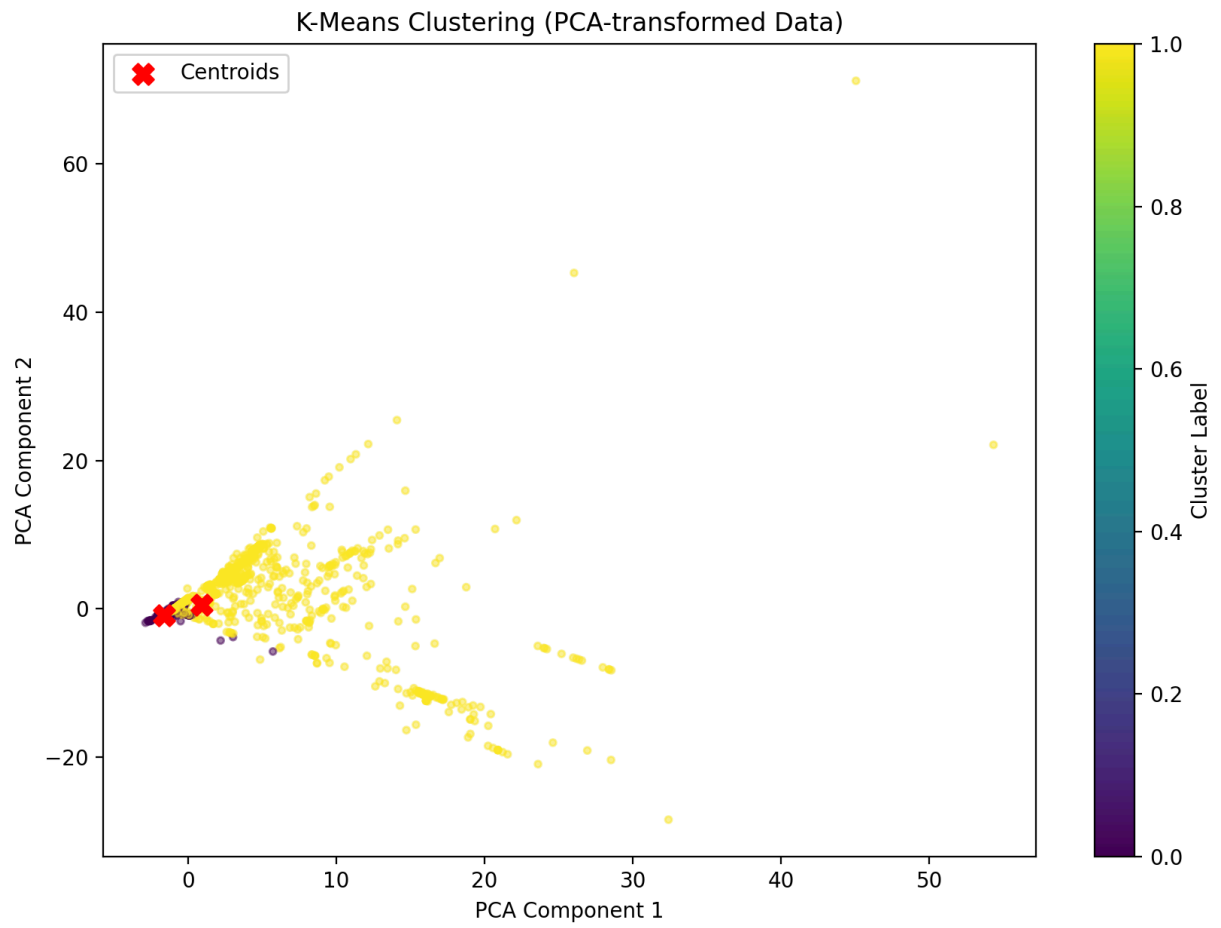
11.4 Table of Summarized Results

Metric	K-Means Clustering	Hierarchical Clustering
Silhouette Score	0.9213	0.6173
Calinski-Harabasz Index	250.87	568.47
Davies-Bouldin Index	0.4230	1.6752
Accuracy	0.9817	0.1329
Precision	0.9649	0.9826
Recall	0.9817	0.1329
F1-Score	0.9732	0.2069
Confusion Matrix	[[3434, 2], [62, 0]]	[[403, 3033], [0, 62]]

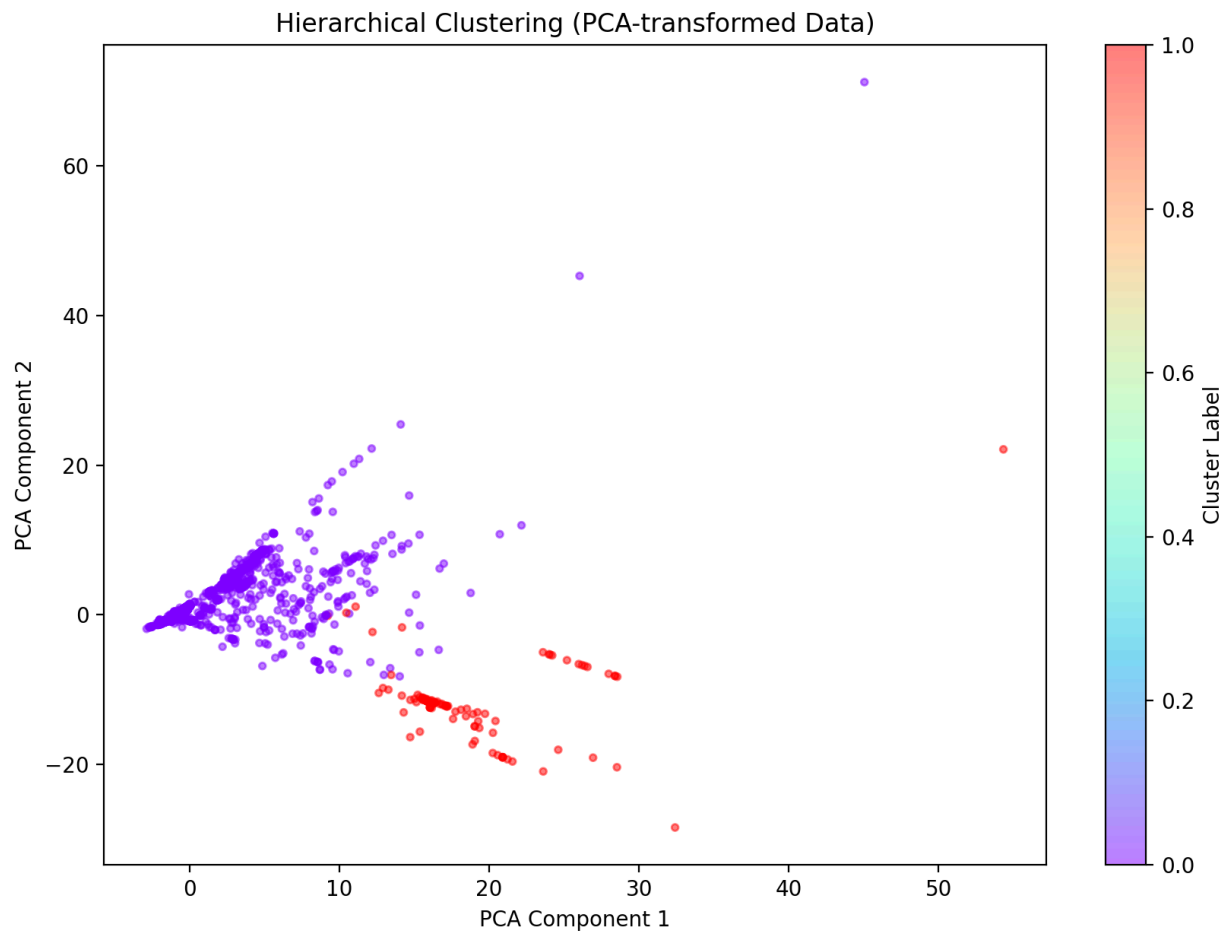
11.5 Dendrogram of Hierarchical Clustering



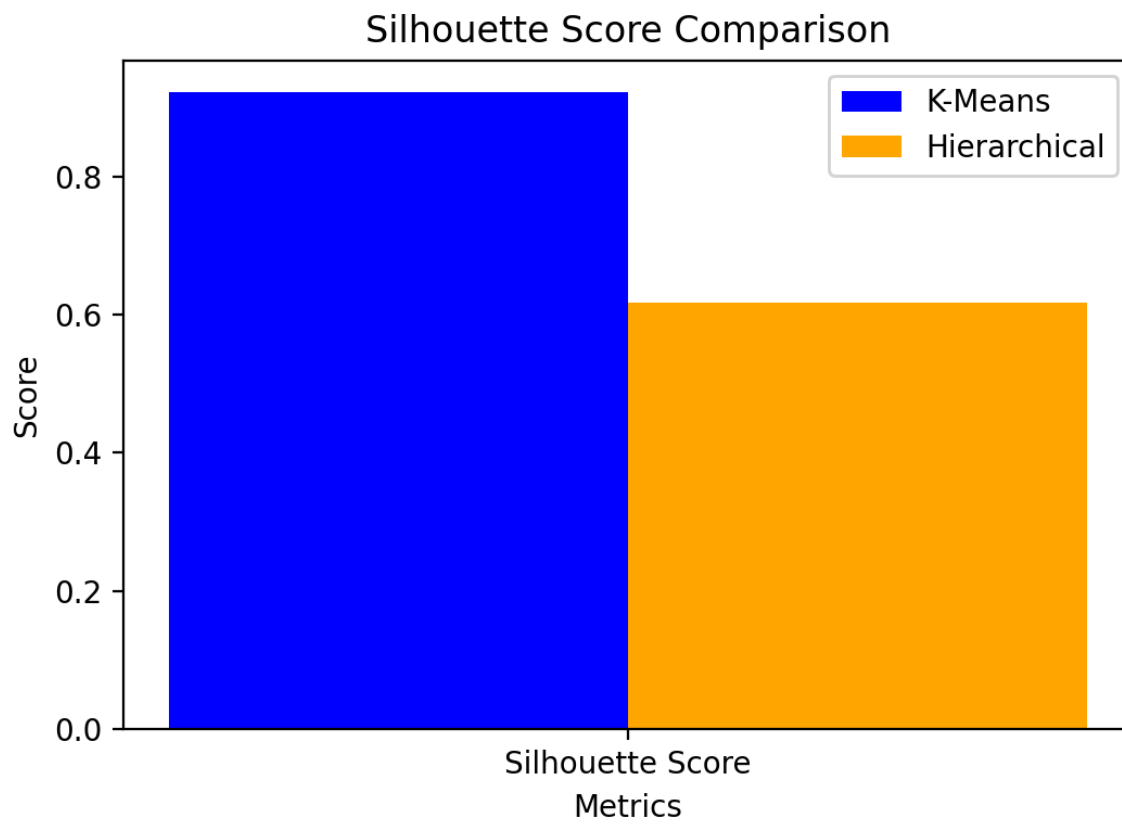
11.6.1 Cluster Scatter Plot for K-Means



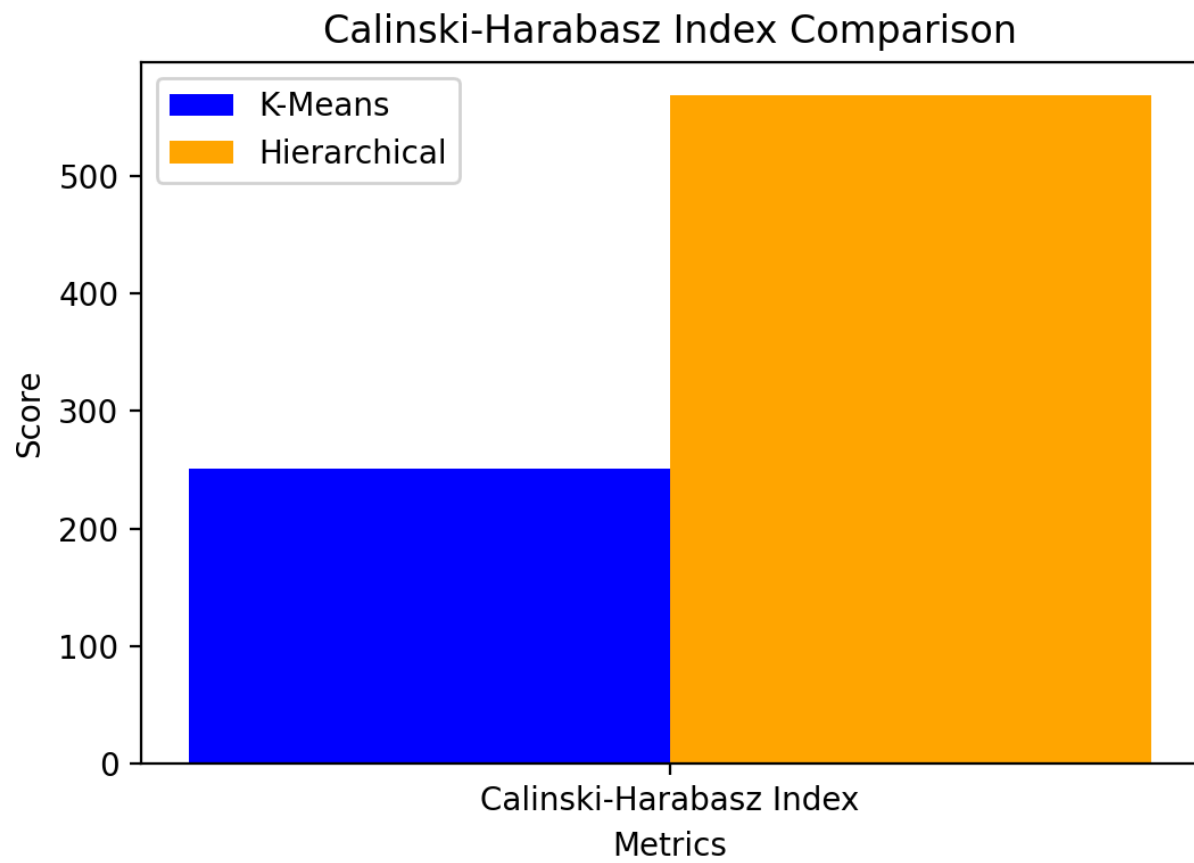
11.6.2 Cluster Scatter Plot for Hierarchical Clustering



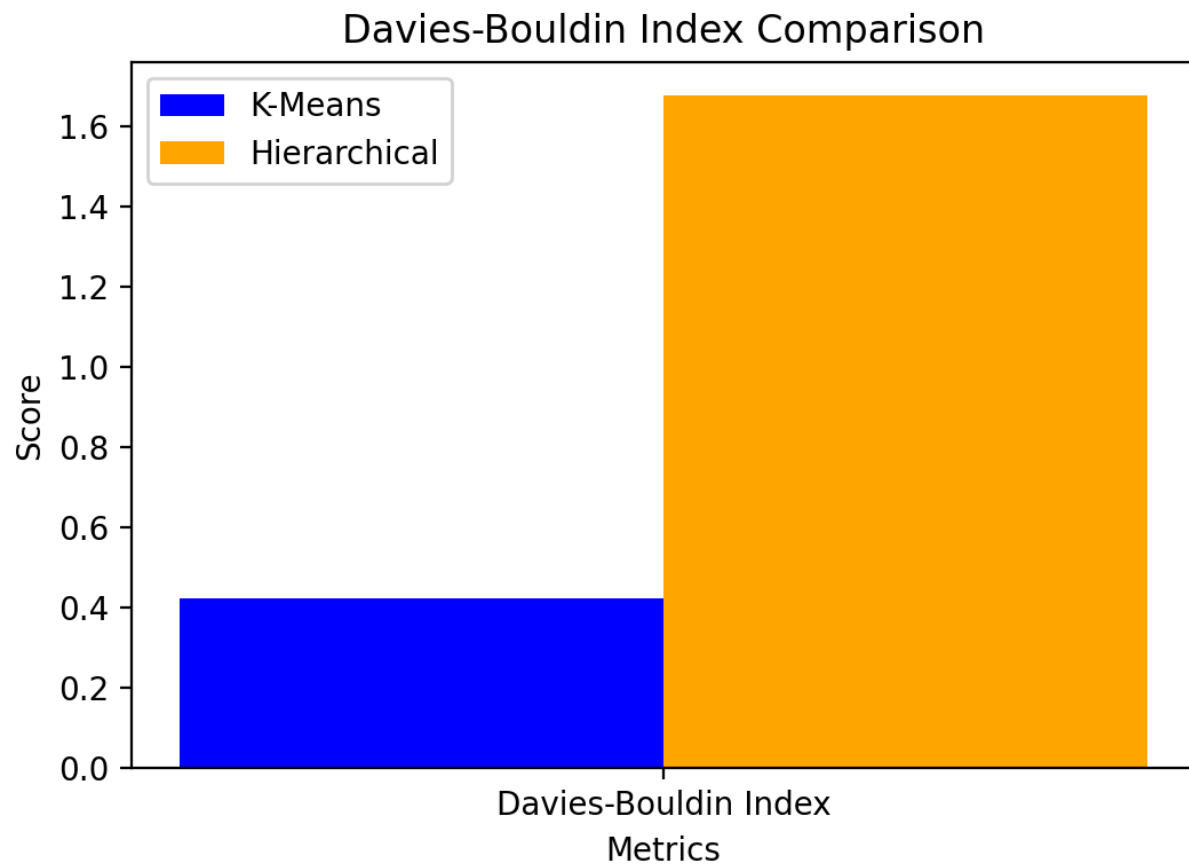
11.7 Bar Graph of Silhouette Score of Both Algorithms



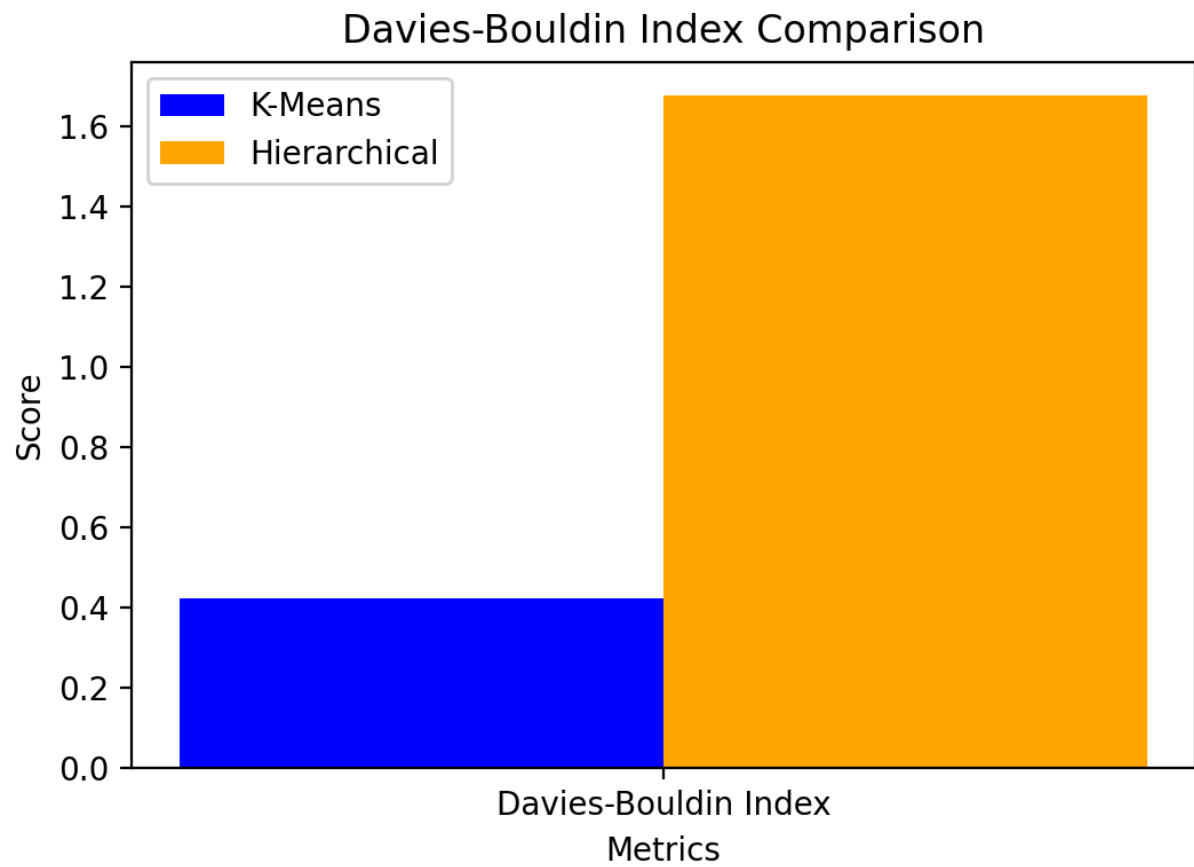
11.8 Bar Graph of Callinski-Harabasz Index of Both Algorithms



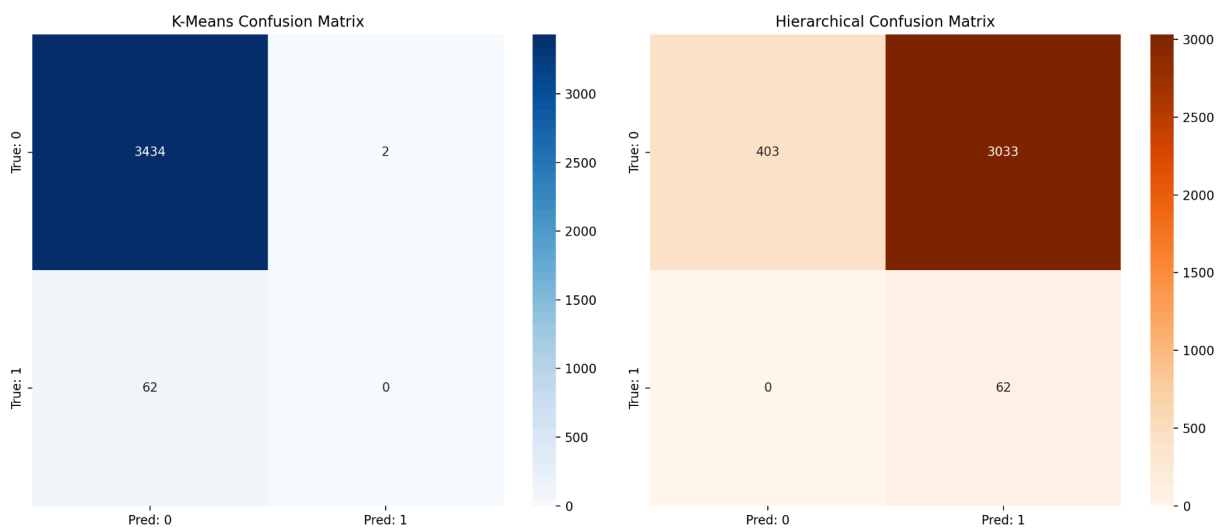
11.9 Bar Graph of Davies-Bouldin Index of Both Algorithms



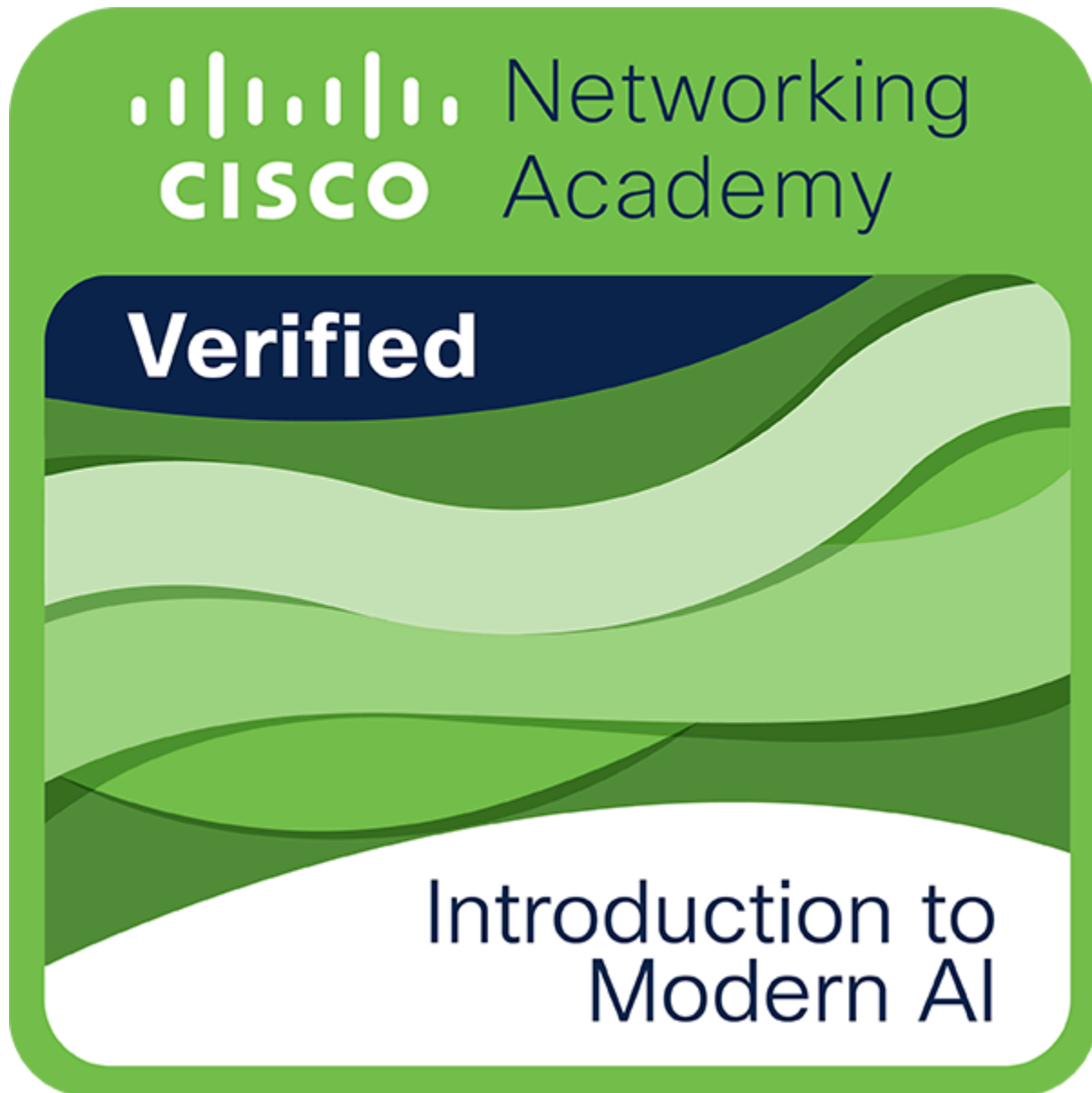
11.10 Bar Graph of Supervised Metrics Comparison



11.11 Confusion Matrix of Both Algorithms



12. CERTIFICATION



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